

CONSTRUCTION (generator -- pre-gate filters)

Held substrate
CRISPRi Perturb-seq
CD4⁺ T cells
(aggregated tables)

Build universe A
KD-gated +
program-significant
= 3,935 genes

Literature-novelty
gate
(rarity + curated
association floor)

**22,039
eligible
(gene, disease)
pairs**

REFEREE (deterministic adjudication -- one verdict + receipt per pair)

Hop 0
Knockdown QC

guide vs. NTC
expression

Hop 1
Effect

on-target DE
(effect size, #DE)

Hop 2
Program

Th1/Th2 shift
(z, adj. p)

Hop 3
Disease

module enrichment
(OR, FDR)

supported

refuted

untested

each pair

experimental receipts (referee-owned)

association receipt
(substrate-inherited stringency)

Cross-vendor independence supplied EXTERNALLY (S4.6): a 5-member replication lab
(3 Opus-class + 2 Codex), 2 clean-room re-implementations -- unanimous pass.

Figure 1. Wayfinder separates construction filters (universe build + novelty gate) from referee decisions (per-pair, per-hop verdicts).